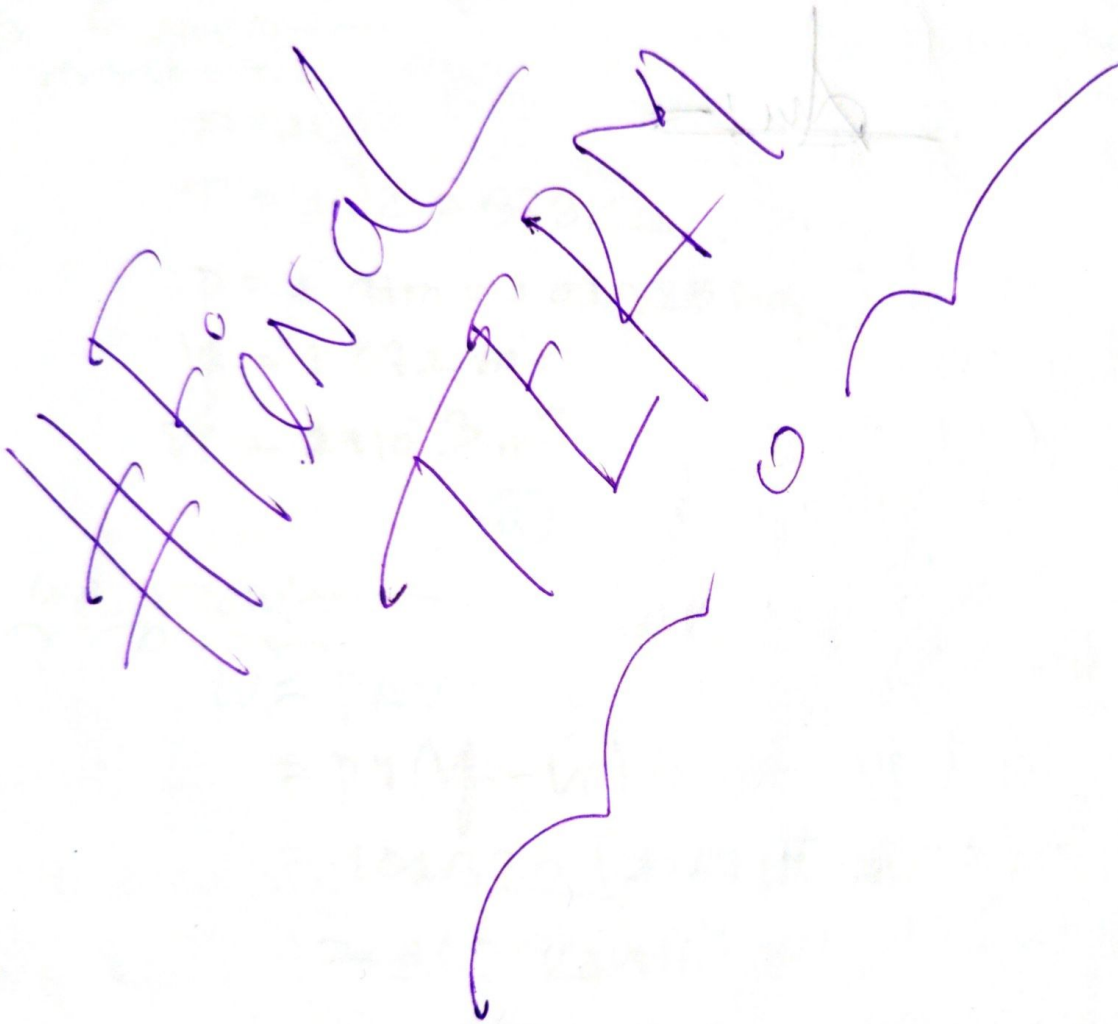


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Mohsin Ibna Hossain
AIUB, Physics 1 final term



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Electric Fields

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Electric Fields —

→ The electric field at a certain point is equal to the electric force per unit charge experienced by a charge at the point.

$$\vec{E} = \frac{\vec{F}}{q_0}$$

SI Unit is $= \text{NC}^{-1}$

* if q_0 charge gives $= F$ force

So, 1 unit $= \frac{F}{q_0}$

Q:- What is 1 unit positive charge?

⇒ 1 unit positive charge is the amount of electric charge carried by a single proton or equivalent to 1.602×10^{-19} coulombs.

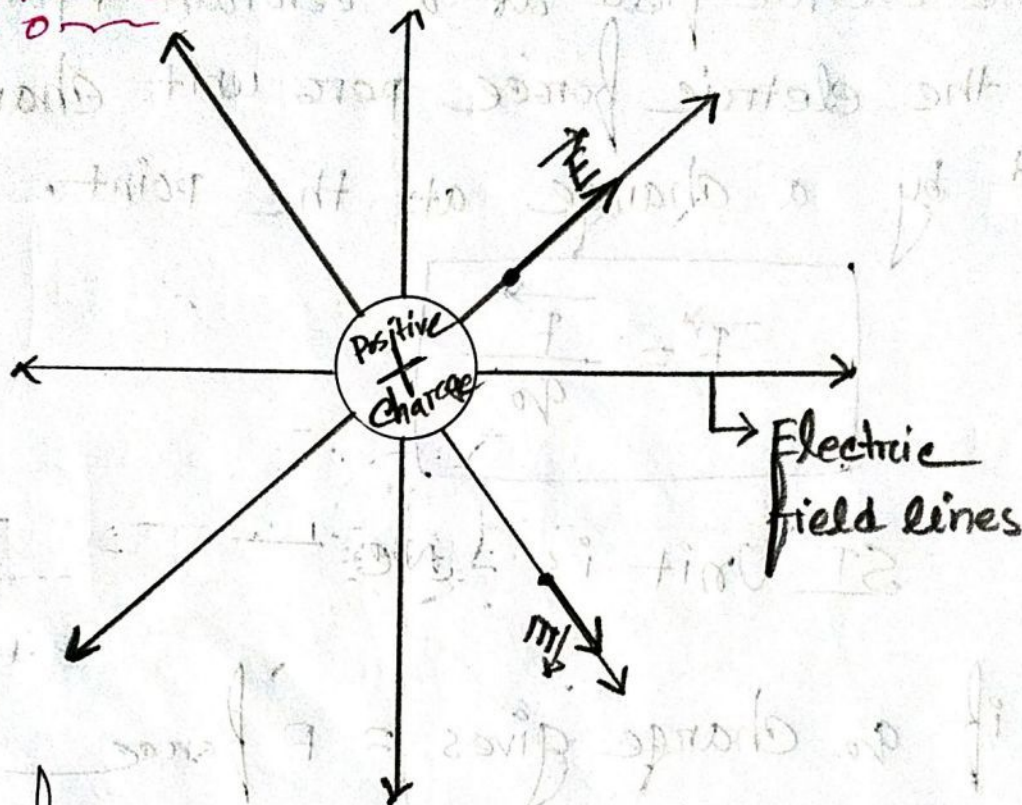
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Electric field lines around charged particles:—

For Positive:—



⇒ The field produced by a positive point charge points away from the charge.

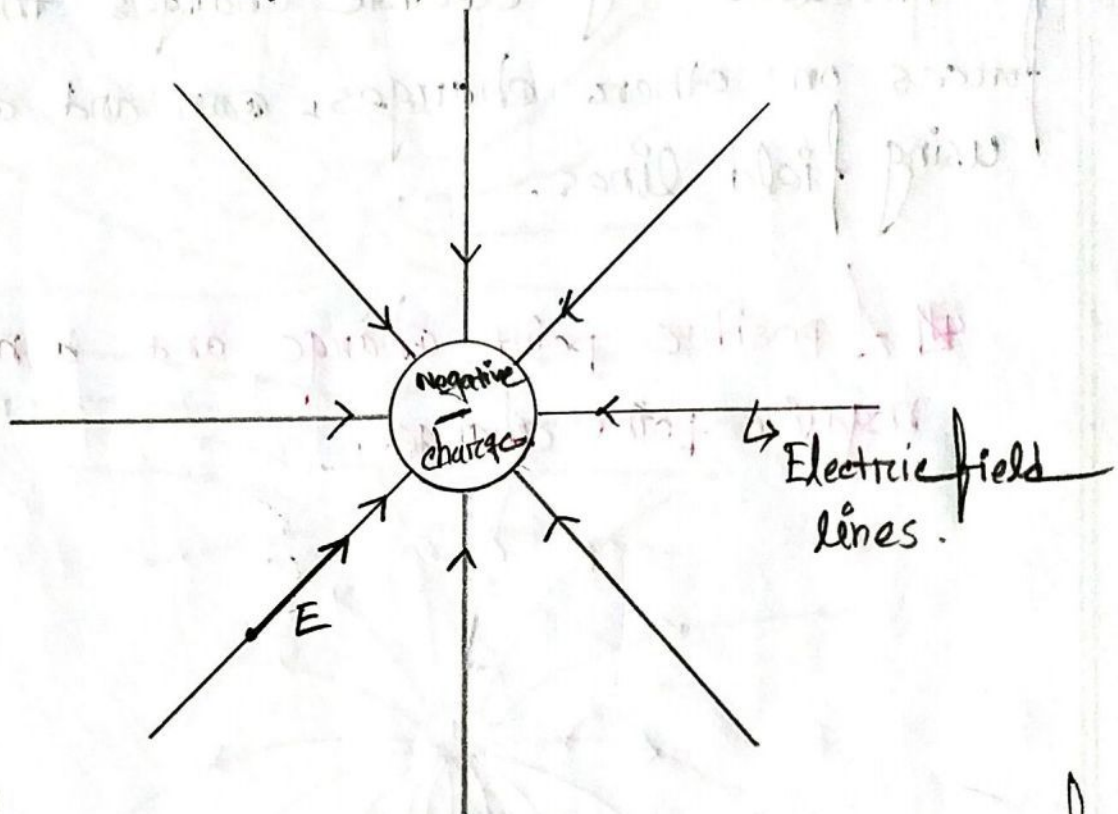
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for negative charges.



⇒ The field produced by a negative point charge. Points toward the charge.

Note

→ Electric field lines extend away from positive charge (where they originate) and toward negative charge (where they terminate).

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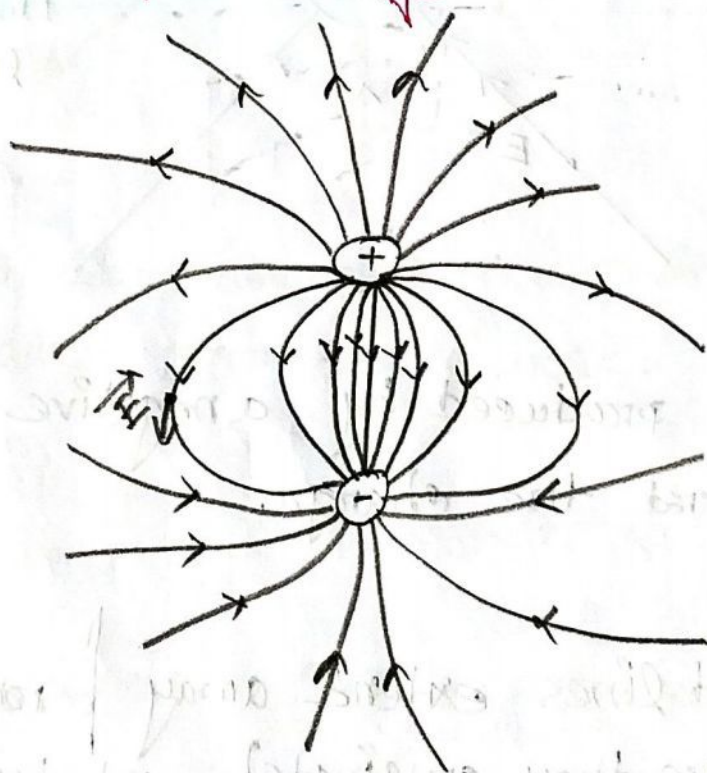
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Properties of Electric field:-

Electric fields are a vector quantities produced by electric charges that exert forces on other charges, and can be visualized using field lines.

A positive point charge and a nearby negative point charge.



→ That are equal in magnitude.

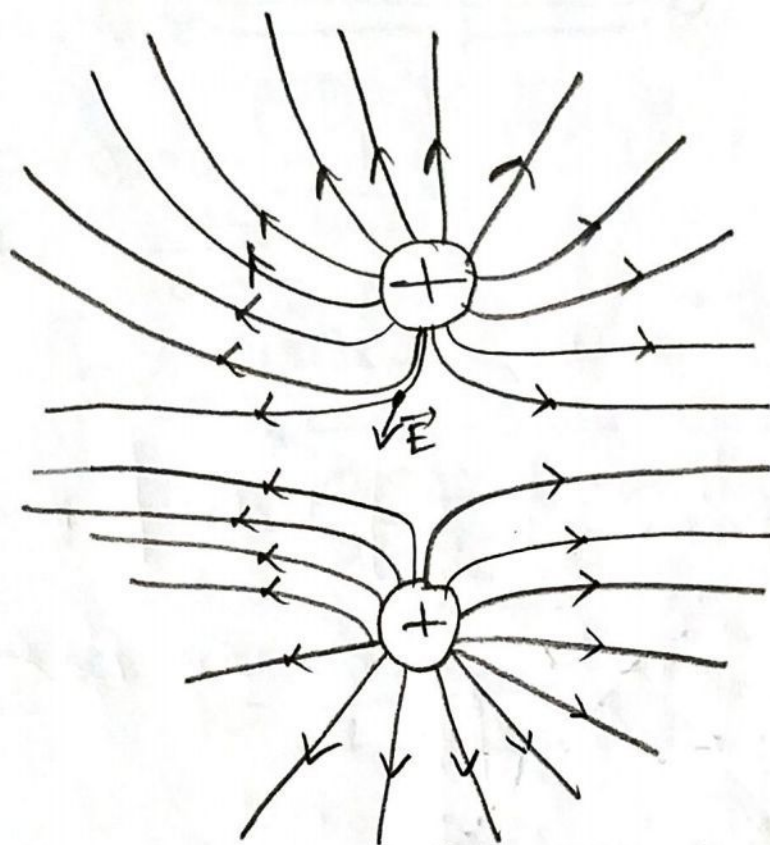
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Two equal positive point charges.



Notes—

→ Electric field lines help us visualize the direction and magnitude of electric field.

The electric field vectors at any point are tangent to the field line through that point. The density of field lines in that region is proportional to the magnitude of the electric field there. Thus, closer field lines represent a strong field.

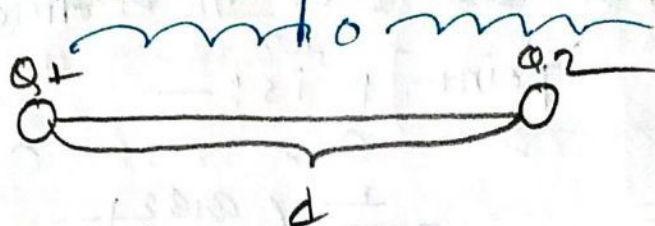
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State of colombs law:-



$$F \propto Q_1 \times Q_2$$

$$F \propto \frac{1}{d^2}$$

So, $F \propto \frac{Q_1 \times Q_2}{d^2}$

$$F = K \frac{Q_1 Q_2}{d^2}$$

where, $K = \frac{1}{4\pi\epsilon_0\epsilon_r}$

now $F = \frac{Q_1 Q_2}{4\pi\epsilon_0\epsilon_r d^2}$

Electric field due to a point charge:-

$$F = \frac{1}{4\pi\epsilon_0} \times \frac{Q_1 Q_2}{r^2}$$

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$$

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the magnitude E of the electric field at point P is: —

$$E = \frac{F}{q_2} = \frac{\frac{1}{4\pi\epsilon_0} \left(\frac{q_1 q_2}{r^2} \right)}{q_2}$$

$$= \frac{1}{4\pi\epsilon_0} \times \frac{q}{r^2}$$

So, $\vec{E} = \frac{1}{4\pi\epsilon_0} \times \frac{q}{r^2} \hat{r}$ → Vector form

$E = \frac{1}{4\pi\epsilon_0} \times \frac{q}{r^2}$ SI Unit NC^{-1}

Q. Problem 50 —

A charged particle produces an electric field with a magnitude of $2NC^{-1}$ at a point that is 50 cm away from the particle. What is the magnitude of the particle's charge?

⇒ is given

$$E = 2NC^{-1}$$

$$r = 50 \times 10^{-2}$$

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$$\frac{1}{4\pi\epsilon_0} = \frac{1}{4\pi \times 8.85 \times 10^{-12}} = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$$

we know,
from

$$E = \frac{1}{4\pi\epsilon_0} \times \frac{q}{r^2}$$

$$\Rightarrow q = \frac{E}{9 \times 10^9} \times r^2$$

$$= \frac{2}{9 \times 10^9} \times (50 \times 10^{-2})^2$$

$$= 5.556 \times 10^{-11} \text{ C}$$

Q:- What is the magnitude of a point charge that would create an electric field of 1.00 N/C at points 1.00 m away?

$\Rightarrow$ is given,
from

$$E = 1 \text{ N/C}$$

$$r = 1 \text{ m}$$

we know,
from

$$E = k \times \frac{q}{r^2}$$

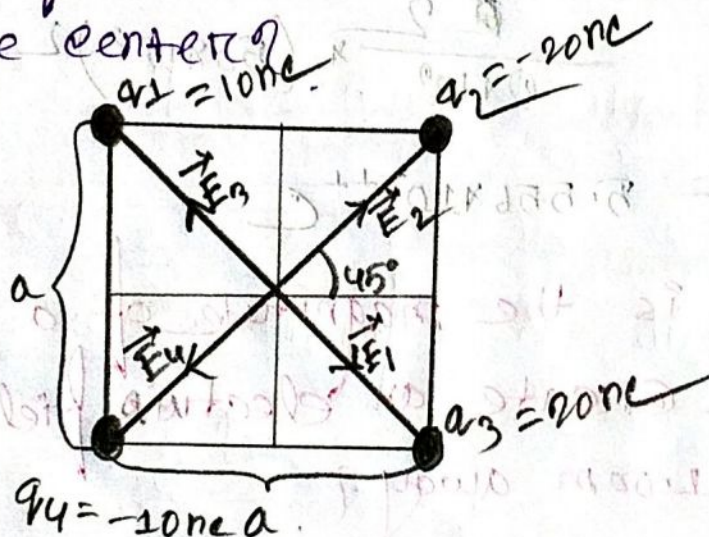
$$\Rightarrow q = \frac{Er^2}{k}$$

$$= \frac{1 \times 1^2}{9 \times 10^9}$$

$$= 1.11 \times 10^{-10} \text{ C}$$

Problem 7

⇒ In the adjacent figure, the four particles form a square of edge length $a = 5.00 \text{ cm}$ and have a charge $q_1 = +1.0 \text{ nC}$, $q_2 = -2.0 \text{ nC}$, $q_3 = +2.0 \text{ nC}$, and $q_4 = -1.0 \text{ nC}$. In unit vector notation, what net electric field do the particles produce at the square center?



The net electric field at the center of the square along the x-axis is :-

$$E_x = E_2 \cos 45^\circ + E_4 \cos 45^\circ - (E_3 \cos 45^\circ + E_1 \cos 45^\circ) \quad \text{--- (i)}$$

hence,

$$E_1 = E_4 = \frac{1}{4\pi\epsilon_0} \left(\frac{1.0 \times 10^{-9}}{r^2} \right) \quad \text{--- (iii)}$$

$$E_2 = E_3 = \frac{1}{4\pi\epsilon_0} \left(\frac{2.0 \times 10^{-9}}{r^2} \right) \quad \text{--- (iv)}$$

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Substitute the value of E_3 and E_4 in equation (I)

$$E_x = E_2 \cos 45^\circ + E_1 \cos 45^\circ - (E_2 \cos 45^\circ + E_1 \cos 45^\circ)$$

$$= E_2 \cos 45^\circ - E_2 \cos 45^\circ + E_1 \cos 45^\circ - E_1 \cos 45^\circ$$

$$= 0$$

The net electric field at the center of the square along y axis is —

$$E_y = E_2 \cos 45^\circ + E_3 \cos 45^\circ - (E_1 \cos 45^\circ + E_4 \cos 45^\circ) \quad \text{--- (II)}$$

Substitute the value of E_3 and E_4 in equation (II) $\Rightarrow$

$$E_y = E_2 \cos 45^\circ + E_2 \cos 45^\circ - (E_1 \cos 45^\circ + E_1 \cos 45^\circ)$$

$$= 2E_2 \cos 45^\circ - 2E_1 \cos 45^\circ$$

$$= 2 \times \frac{1}{4\pi\epsilon_0} \times \frac{q_2}{r^2} \cos 45^\circ - 2 \times \frac{1}{4\pi\epsilon_0} \times \frac{q_1}{r^2} \cos 45^\circ$$

$$= 2 \times \frac{1}{4\pi\epsilon_0} \times \frac{1}{r^2} \cos 45^\circ (q_2 - q_1)$$

$$= 2 \times 9 \times 10^9 \times \frac{1}{(0.0354)^2} \times \cos 45^\circ (2 \times 10^{-9} - 1 \times 10^{-9})$$

$$= 101.566 \times 10^3 \text{ N/C}$$

$$\text{Total } E = E_x \hat{i} + E_y \hat{j} = 0 + 101.566 \times 10^3 \hat{j}$$

$$= 101.566 \times 10^3 \hat{j} \text{ N/C}$$

$$\begin{aligned} q_p^2 &= a^2 + a^2 \\ q_p &= \sqrt{a^2 + a^2} \\ q_p &= \sqrt{2a^2} \\ OP &= \frac{\sqrt{2 \times (0.0354)^2}}{2} \end{aligned}$$

$$= 0.0354$$

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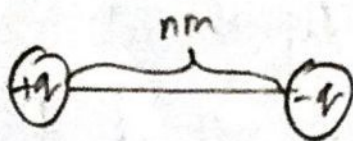
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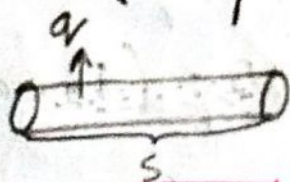
Dipoles

⇒ A dipole is a pair of equal and opposite electric charge or magnetic poles separated by a small distance.



Linear Charge density

⇒ Linear Charge density is the amount of electric charge per unit length of a one dimensional object.



$$\lambda = \frac{q}{s}$$

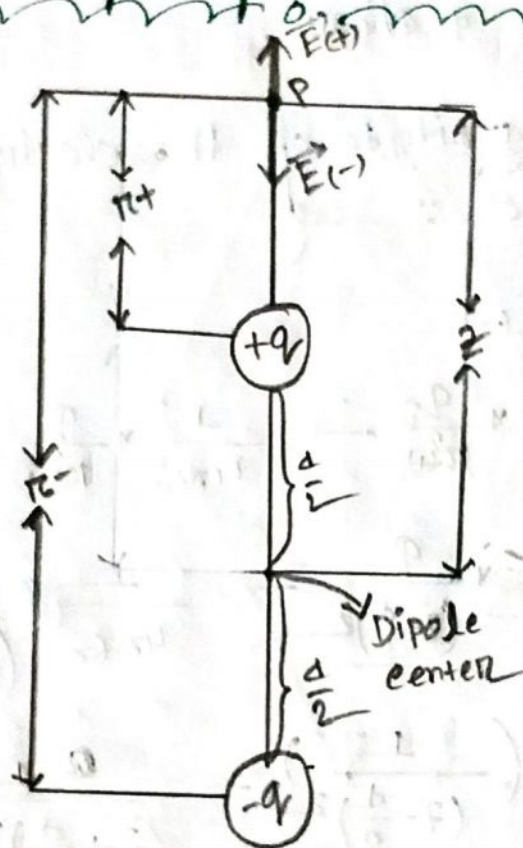
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Electric fields—

Electric field for a electric Dipole:—



We all know the normal equation is—

$$E = \frac{1}{4\pi\epsilon_0} \times \frac{q}{r^2}$$

→ This is the normal equation for a single charge. But for a dipole, we need a different equation.

→ Now, let's consider a double charge system. The equation for the electric field of a dipole is—

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$$r_{b+} = z - \frac{d}{2}$$

$$r_{b-} = z + \frac{d}{2}$$

& $E_+ \rightarrow$ for $+q$ charge.
 $E_- \rightarrow$ for $-q$ charge.

So, the net magnitude of the electric field at point P is—

$$E = E_+ - E_-$$

$$= \frac{1}{4\pi\epsilon_0} \times \frac{q}{r_{b+}^2} - \frac{1}{4\pi\epsilon_0} \times \frac{q}{r_{b-}^2}$$

$$= \frac{1}{4\pi\epsilon_0} \times \frac{q}{\left(z - \frac{d}{2}\right)^2} - \frac{1}{4\pi\epsilon_0} \times \frac{q}{\left(z + \frac{d}{2}\right)^2}$$

$$= \frac{q}{4\pi\epsilon_0} \left\{ \frac{1}{\left(z - \frac{d}{2}\right)^2} - \frac{1}{\left(z + \frac{d}{2}\right)^2} \right\}$$

$$= \frac{q}{4\pi\epsilon_0} \left\{ \frac{1}{z^2 \left(1 - \frac{d}{2z}\right)^2} - \frac{1}{z^2 \left(1 + \frac{d}{2z}\right)^2} \right\}$$

$$= \frac{q}{4\pi\epsilon_0} \times \frac{1}{z^2} \left\{ \frac{1}{\left(1 - \frac{d}{2z}\right)^2} - \frac{1}{\left(1 + \frac{d}{2z}\right)^2} \right\}$$

$$= \frac{1}{4\pi\epsilon_0} \times \frac{q}{z^2} \left\{ \left(1 - \frac{d}{2z}\right)^{-2} - \left(1 + \frac{d}{2z}\right)^{-2} \right\}$$

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Here $\frac{d}{2z} \ll 1$

$$\text{So, } E = \frac{1}{4\pi\epsilon_0} \times \frac{q}{z^2} \left[1 - (-2) \frac{d}{2z} - \left(1 + (-2) \frac{d}{2z} \right) \right]$$

$$E = \frac{1}{4\pi\epsilon_0} \times \frac{q}{z^2} \left[1 + \frac{2d}{2z} - 1 + \frac{2d}{2z} \right]$$

$$E = \frac{1}{4\pi\epsilon_0} \times \frac{q}{z^2} \left[2 \times \frac{d}{z} \right]$$

$$E = \frac{1}{4\pi\epsilon_0} \times \frac{q}{z^2} \times \frac{2d}{z}$$

$$E = \frac{q d}{4\pi\epsilon_0 z^3}$$

$$E = \frac{q d}{2\pi\epsilon_0 z^3}$$

$$\text{So, } E = \frac{q d}{2\pi\epsilon_0 z^3}$$

$$E = \frac{P}{2\pi\epsilon_0 z^3}$$

if $d \ll z$ then

$$\frac{d}{2z} \ll 1$$

Binomial theorem —

$$(1+x)^n = 1 + \frac{nx}{1!} + \frac{n(n-1)x^2}{2!} + \dots (n \geq 1)$$

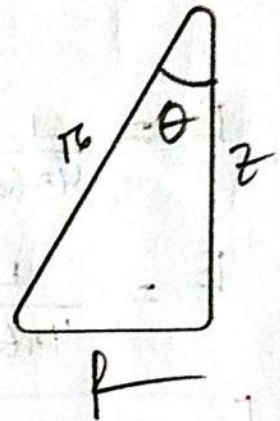
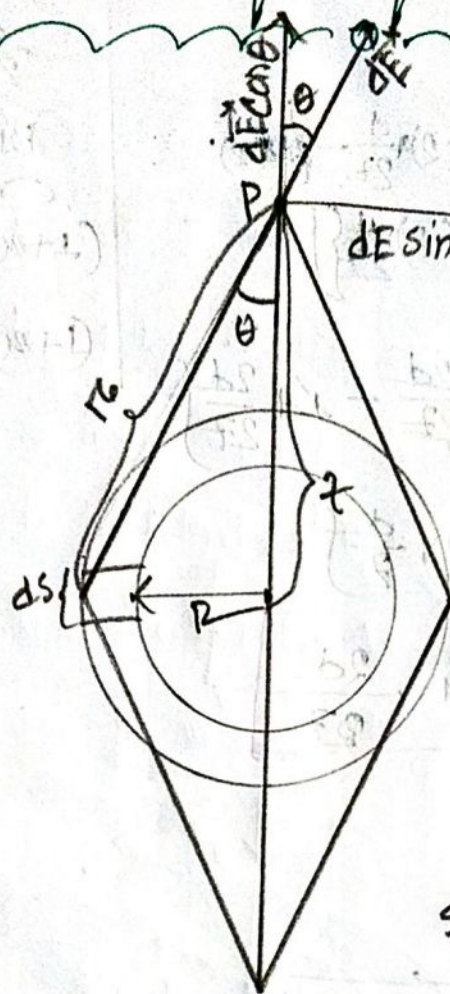
$$(1+x)^n \approx 1 + nx \text{ for } n \ll 1$$

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Electric field for A ring:-



$$\cos \theta = \frac{z}{r}$$

$$r = \sqrt{R^2 + z^2}$$

$$\text{So, } \cos \theta = \frac{z}{\sqrt{R^2 + z^2}}$$

$$dF = \frac{1}{4\pi\epsilon_0} \times \frac{dq}{r^2}$$

$$= \frac{1}{4\pi\epsilon_0} \times \frac{\lambda ds}{r^2}$$

$$= \frac{1}{4\pi\epsilon_0} \times \frac{\lambda ds}{(\sqrt{R^2 + z^2})^2}$$

for linear charge density $\lambda = \frac{dq}{ds}$

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$$\text{So, } dE \cos \theta = \frac{1}{4\pi\epsilon_0} \times \frac{\lambda ds}{(\sqrt{r^2+z^2})^2} \times \frac{z}{\sqrt{r^2+z^2}}$$

$$E_z = \int_0^{2\pi R} dE \cos \theta = \int_0^{2\pi R} \frac{1}{4\pi\epsilon_0} \times \frac{\lambda ds}{(\sqrt{r^2+z^2})^2} \times \frac{z}{\sqrt{r^2+z^2}}$$

$$= \frac{1}{4\pi\epsilon_0} \times \frac{\lambda z}{(\sqrt{r^2+z^2})^3} \times \frac{1}{(r^2+z^2)^{\frac{1}{2}}} \int_0^{2\pi R} ds$$

$$= \frac{\lambda z}{4\pi\epsilon_0 (r^2+z^2)^{\frac{3}{2}}} \int_0^{2\pi R} ds$$

$$= \frac{\lambda z}{4\pi\epsilon_0 (r^2+z^2)^{\frac{3}{2}}} [s]_0^{2\pi R}$$

$$= \frac{\lambda z \times 2\pi R}{4\pi\epsilon_0 (r^2+z^2)^{\frac{3}{2}}}$$

$$E_z = \frac{qz}{4\pi\epsilon_0 (r^2+z^2)^{\frac{3}{2}}}$$

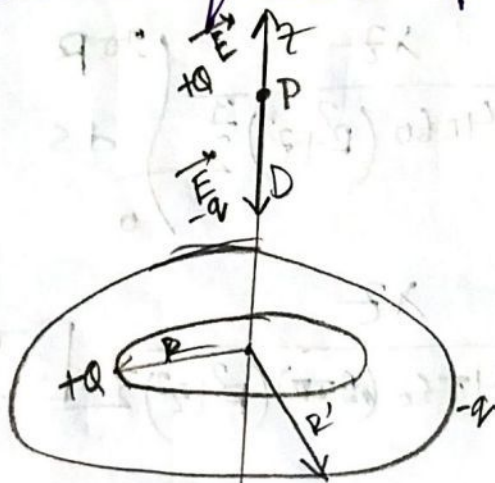
$$\lambda = \frac{dq}{ds}$$

$$dq = \lambda ds$$

$$q = \lambda \times 2\pi R$$

Problem 30:-

Figure shows two concentric rings, of radii R and $R' = 3R$, that lie on the same plane. Point P lies on the central z axis, at distance $D = 2R$ from the center of the rings. The smaller ring has uniformly distributed charge $+q$. In terms of q , what is the uniformly distributed charge on the larger ring if the net electric field at P is zero.



⇒ According to the statement of the problem,

$$E_{+q} - E_{-q} = 0$$

$$E_{+q} = E_{-q} \quad \text{--- (1)}$$

we know,

$$E = \frac{qz}{4\pi\epsilon_0(z^2 + R^2)^{3/2}}$$

$$\left\{ \begin{array}{l} D = 2R \\ \text{1st case: } R = R \\ \text{2nd } R' = 3R \end{array} \right.$$

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Substitute the Value of E in equation ①

$$\Rightarrow \frac{QD}{4\pi\epsilon_0 (D^2 + R^2)^{\frac{3}{2}}} = \frac{QD}{4\pi\epsilon_0 (D^2 + R^2)^{\frac{3}{2}}}$$

$$\Rightarrow \frac{Q}{(D^2 + R^2)^{\frac{3}{2}}} = \frac{Q}{(D^2 + (3R)^2)^{\frac{3}{2}}}$$

$$\Rightarrow \frac{Q}{Q} = \frac{(D^2 + R^2)^{\frac{3}{2}}}{(D^2 + 9R^2)^{\frac{3}{2}}}$$

$$\Rightarrow \frac{Q}{Q} = \frac{(4R^2 + R^2)^{\frac{3}{2}}}{(4R^2 + 9R^2)^{\frac{3}{2}}}$$

$$\Rightarrow \frac{Q}{Q} = \left(\frac{5R^2}{13R^2} \right)^{\frac{3}{2}}$$

$$\Rightarrow \frac{Q}{Q} = \left(\frac{5}{13} \right)^{\frac{3}{2}}$$

$$\Rightarrow \frac{Q}{Q} = 0.23$$

$$\Rightarrow \frac{Q}{Q} = \frac{1}{0.23}$$

$$\Rightarrow Q = 4.10Q$$

$$Q = -4.10Q$$

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Q:- What is electric Flux

⇒ Electric flux is the measure of the total numbers of electric field that pass through a particular surface.

Electric Flux for uniform $\vec{E}$ and through a flat surface (open plane):-

⇒ The electric flux Φ through the flat surface equals the scalar product of the electric field $\vec{E}$ and the area vector $\vec{A}$

$$\Phi = \vec{E} \cdot \vec{A}$$

$$\Phi = EA \cos \theta$$

$$\Phi_{\max} = EA \cos 0^\circ$$

$$\Phi_{\min} = EA \cos 90^\circ$$

-98-200 / SI Unit = $\text{N-m}^2/\text{C}$

P.T.O

Q. Write down the condition of electric flux - 2.

1. electric flux maximum; when $\theta = 0^\circ$

2. electric flux minimum; when $\theta = 90^\circ$

electric flux for nonuniform $\vec{E}$

And through a curved surface:-

⇒ When the surface is curved and field is nonuniform, we calculate the flux by dividing the surface into small patches dA , so that each differential area is essentially flat and field is uniform over each patch.

$$\Phi = \int \vec{E} \cdot d\vec{A}$$

Electric Flux through a closed surface:—

⇒ The net flux through a closed surface is given by

$$\Phi = \oint \vec{E} \cdot d\vec{A}$$

$$\Phi = \int \vec{E} \cdot d\vec{A} + \int \vec{E} \cdot d\vec{A} + \int \vec{E} \cdot d\vec{A}$$

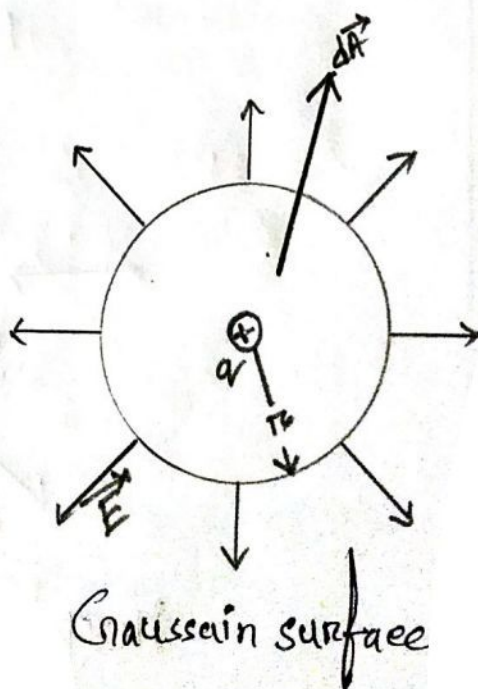
Gauss's Law:—

⇒ Gauss's Law states that the total electric flux (Φ) through any closed surface, known as gaussian surface, is proportional to the total (net) electric charge (q) inside the surface.

$$\Phi = \frac{q}{\epsilon_0}$$

$$\Phi \epsilon_0 = q$$

$$\epsilon_0 \oint \vec{E} \cdot d\vec{A} = q$$

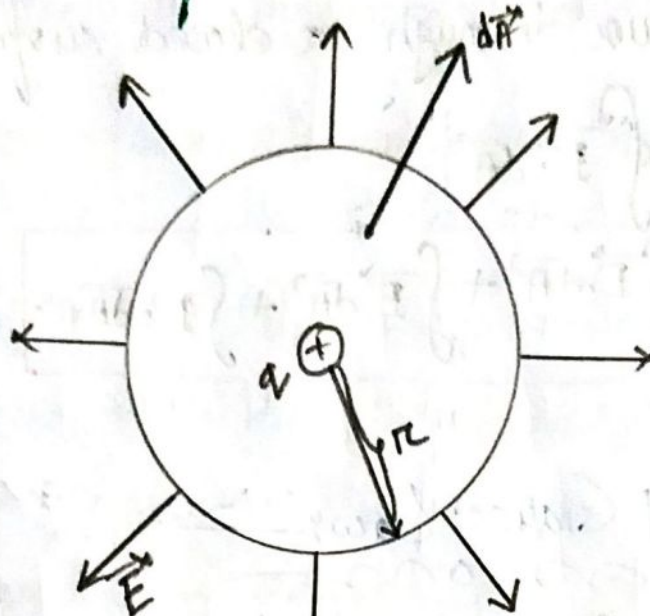


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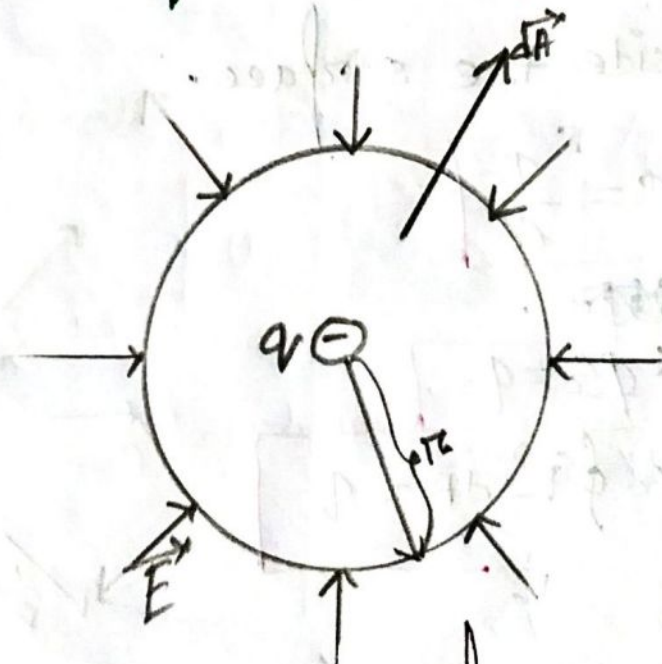
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☒ Gaussian surface around positive charges:-



Positive (outward) flux

☒ Gaussian surface around negative charge:-



Negative (inward) flux

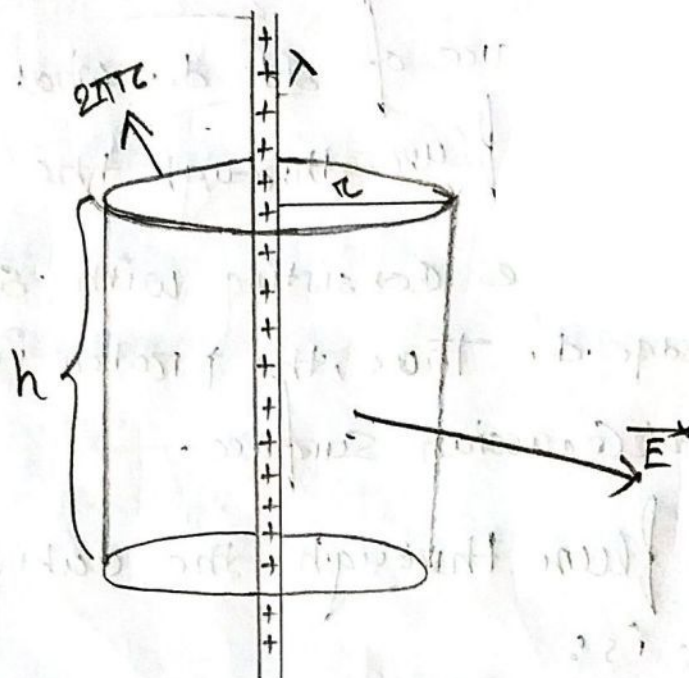
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Applying Gauss' Law: Cylindrical Symmetry:-



⇒ An infinitely long cylindrical plastic rod with a uniform charge density λ

$$\text{radius} = r_0 \quad / \quad A = 2\pi r_0 h$$

$$\text{length} = h$$

Using Gauss' law, we can write...

$$\epsilon_0 \Phi = q = \lambda h$$

$$\epsilon_0 \Phi = \lambda h$$

$$\epsilon_0 E A = \lambda h$$

$$\epsilon_0 E (2\pi r h) = \lambda h$$

$$E = \frac{\lambda}{\epsilon_0 2\pi r}$$

where,

λ = the linear charge density.

Problem 58

⇒ A proton is a distance $\frac{d}{2}$ directly above the center of a square of side d . What is the magnitude of the electric flux through the square?

⇒ Consider a cube consisting with six square faces; each of edge d . Thus, the proton is enclosed by the cubical Gaussian surface.

⇒ Electric flux through the cubical Gaussian surface is:

$$\Phi = \frac{Q}{\epsilon_0} = \frac{e^+}{\epsilon_0} = \frac{1.6 \times 10^{-19}}{8.854 \times 10^{-12}}$$

$$= 180 \times 10^{-9} \text{ N.m}^2/\text{C}$$

So, Electric flux through a face of a cube is,

$$\Phi_s = \frac{180 \times 10^{-9}}{6} = 30 \times 10^{-9} \frac{\text{N.m}^2}{\text{C}}$$

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Problem 25

⇒ An infinite line of charge produces a field of magnitude $4.5 \times 10^4 \text{ N/C}$ at distance 2 m .

Find the linear charge density?

⇒ is given—

$$E = 4.5 \times 10^4 \text{ N/C}$$

$$r = 2 \text{ m}$$

$$\epsilon_0 = 8.85 \times 10^{-12}$$

we know—

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

$$\lambda = E \times 2\pi\epsilon_0 r$$

$$= 4.5 \times 10^4 \times 2\pi \times 8.85 \times 10^{-12}$$

$$= 2.5 \times 10^{-6} \text{ C/m}$$

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Electric Potential

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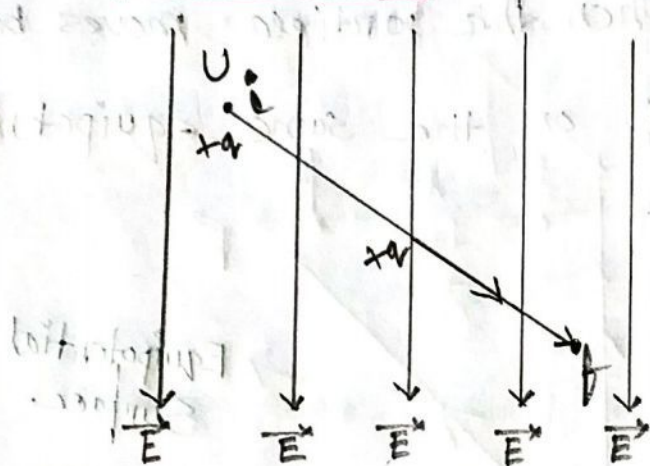
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#Electric potential

⇒ The potential energy (U) per unit charge (q) at a point in a electric field ($\vec{E}$) is called the electric potential.

$$V = \frac{U}{q}$$



the difference in potential energy per unit charge between the two points —

$$\Delta V = V_f - V_i$$

$$\Delta V = \frac{U_f}{q} - \frac{U_i}{q}$$

$$\Delta V = \frac{\Delta U}{q}$$

we can define the potential difference between points i and f as

$$\Delta V = V_f - V_i = \frac{-W}{q}$$

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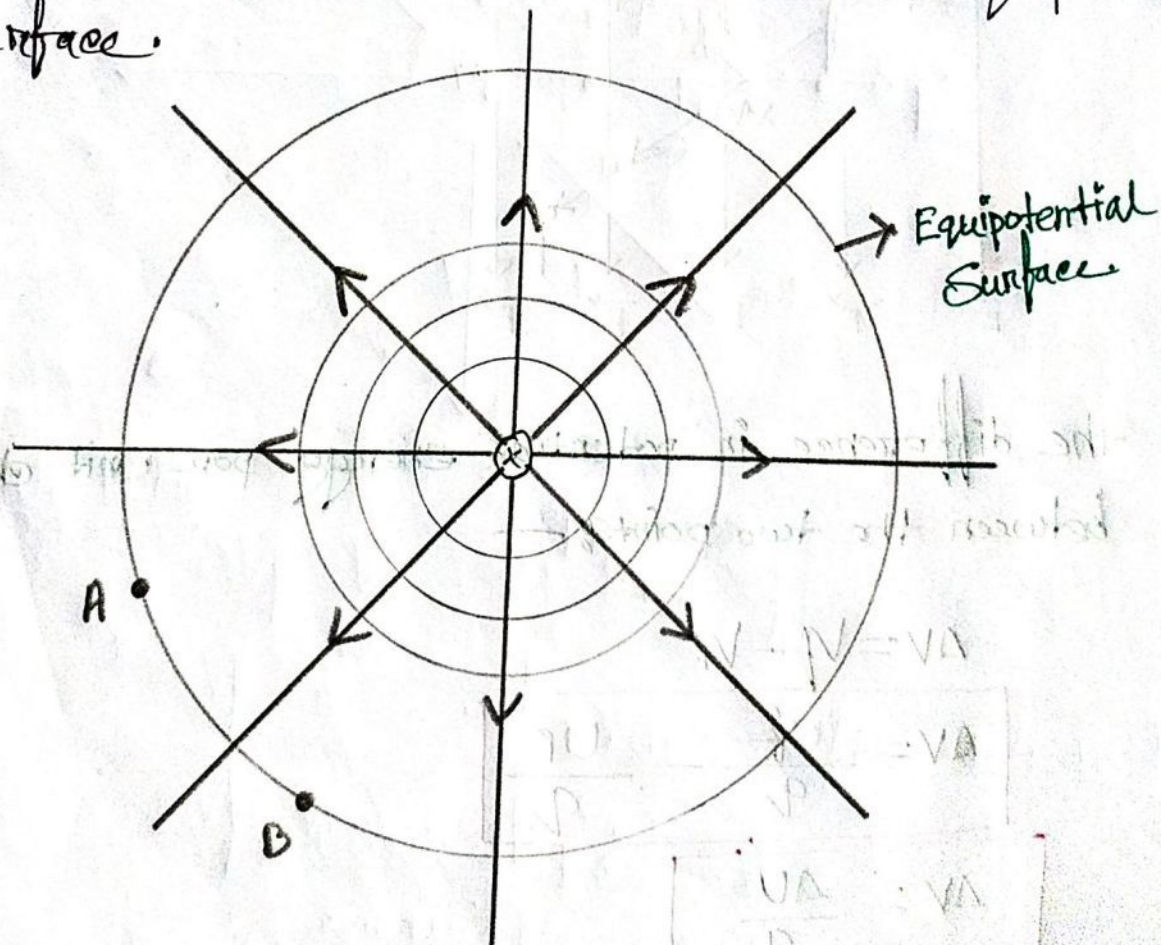
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Equipotential Surfaces

⇒ Adjacent points that have the same electric potential form an equipotential surface, which can be either an imaginary surface or real, physical surface. No net work W is done on a charged particle by an electric field when the particle moves between two points A and B on the same equipotential surface.



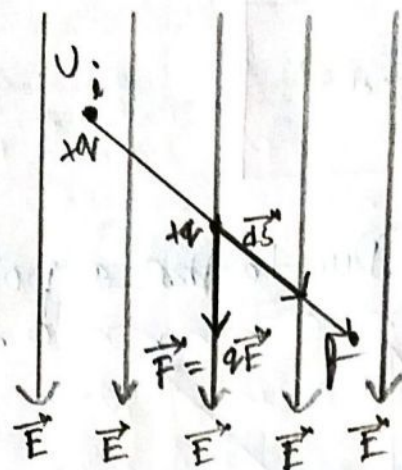
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Calculating the potential from the fields



we know,

$$dW = \vec{F} \cdot d\vec{s}$$

$$dW = q\vec{E} \cdot d\vec{s}$$

The total work W done on the particle by the field as the particle moves from point i to point f ,

$$W = \int_i^f dW = \int_i^f q\vec{E} \cdot d\vec{s} = q \int_i^f \vec{E} \cdot d\vec{s}$$

So, the work done by electrostatic force in terms of potential difference:

$$\Delta V = V_f - V_i = - \frac{W}{q}$$

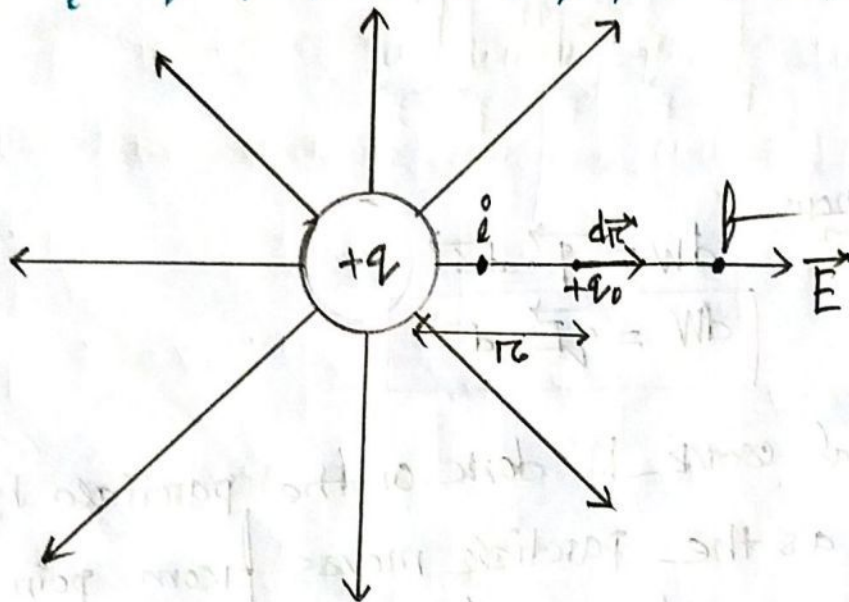
$$V_f - V_i = \frac{-q \int_i^f \vec{E} \cdot d\vec{s}}{q}$$

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$$V_f - V_i = - \int_i^f \vec{E} \cdot d\vec{s}$$

$$V_f - V_i = - \int_i^f \vec{E} \cdot d\vec{s}$$

potential Due to a point charge:-



We know,

$$V_f - V_i = - \int_i^f \vec{E} \cdot d\vec{s}$$

$$V_f - V_i = - \int_i^f \vec{E} \cdot d\vec{r} \quad [\text{here } d\vec{s} = d\vec{r}]$$

$$V_f - V_i = - \int_i^f E \cos 0^\circ \cdot dr$$

$$= - \int_i^f E dr$$

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$$V_f - V_i = - \frac{q}{4\pi\epsilon_0} \int_i^f \frac{1}{r^2} dr$$

$$V_f - V_i = - \frac{q}{4\pi\epsilon_0} \left(- \left| \frac{1}{r} \right|_{r=r_i}^{r=r_f} \right)$$

$$V_f - V_i = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{r_f} - \frac{1}{r_i} \right]$$

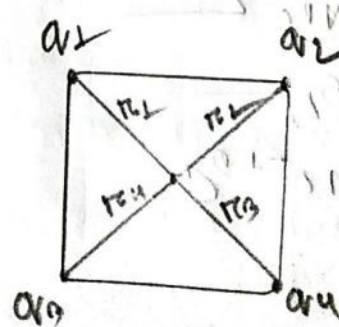
if $r_f = \infty$ then $V_f = 0$ and $\frac{1}{r_f} = 0$

$$\therefore 0 - V_i = \frac{q}{4\pi\epsilon_0} \left(0 - \frac{1}{r_i} \right)$$

$$-V_i = - \frac{q}{4\pi\epsilon_0} \left[\frac{1}{r_i} \right]$$

$$V = \frac{1}{4\pi\epsilon_0} \times \frac{q}{r}$$

Potential Due to the Group of point charges: _____



$$V = \sum_{i=1}^n V_i = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_i}$$

$$V = \sum_{i=1}^4 V_i = \frac{1}{4\pi\epsilon_0} \left[\frac{q_1}{r_1} + \frac{q_2}{r_2} + \frac{q_3}{r_3} + \frac{q_4}{r_4} \right]$$

Q: 4:-

⇒ Two large, parallel conducting plates are 12 cm apart and have a charge of equal magnitude and opposite sign on their facing surfaces. An electric force of $3.9 \times 10^{-15} \text{ N}$ acts on an electron placed anywhere between the two plates.

① Find the electric field at the position of the electron.

② What is the potential difference between the plates?

⇒ is given,

$$d = 12 \text{ cm} = 12 \times 10^{-2} \text{ m}$$

$$F = 3.9 \times 10^{-15} \text{ N}$$

$$e^- = q = 1.6 \times 10^{-19} \text{ C}$$

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(a)

We know,

$$F = qE$$

$$E = \frac{F}{q}$$

$$= \frac{3.94 \times 10^{-15}}{1.6 \times 10^{-19}}$$

$$= 2.4375 \times 10^4 \text{ N/C}$$

(b)

We know:—

$$|AV| = Ed$$

$$= 2.4375 \times 10^4 \times 12 \times 10^{-2}$$

$$= 2925 \text{ Volt}$$

Problem 86:—

When an electron moves from A to B along an electric field line in the adjacent figure, the electric field does $3.94 \times 10^{-15} \text{ J}$ of work on it. What are the electric potential differences

(a) $V_B - V_A$

(b) $V_C - V_A$

(c) $V_C - V_B$

$\Rightarrow$ is given,

$$W = 3.04 \times 10^{-19} \text{ J}$$

$$q = 1.6 \times 10^{-19} \text{ C}$$

(a)

$$\begin{aligned} V_B - V_A &= \frac{-W}{q} \\ &= \frac{-3.04 \times 10^{-19}}{-1.6 \times 10^{-19}} \text{ Volt} \\ &= 2.4625 \text{ Volt} \end{aligned}$$

(b)

$$\begin{aligned} V_C - V_A &= \frac{-W}{q} \\ &= \frac{-3.04 \times 10^{-19}}{-1.6 \times 10^{-19}} \text{ Volt} \\ &= 2.4625 \text{ Volt} \end{aligned}$$

(c)

$$V_C - V_B = 0 \text{ Volt}$$

Because points B and C are the same equipotential surface.

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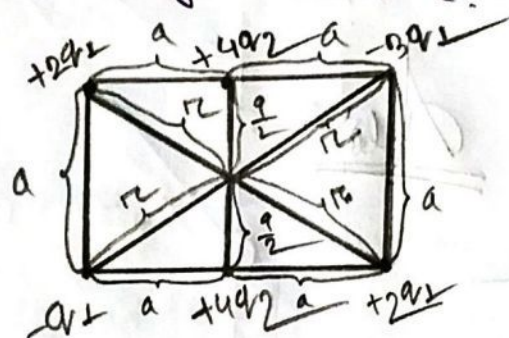
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Problem 16: —

Figure shows a rectangular array of charged particles fixed in place, with distance $a = 30 \text{ cm}$, and the charge shown as integer multiples of $q_1 = 3.4 \text{ pC}$ and $q_2 = 6 \text{ pC}$. With $V = 0$ at infinity, what is the net electric potential at the rectangle's center?



is given: —

$$a = 30 \text{ cm} = 0.30 \text{ m}$$

$$q_1 = 3.40 \text{ pC} = 3.40 \times 10^{-12} \text{ C}$$

$$q_2 = 6 \text{ pC} = 6.0 \times 10^{-12} \text{ C}$$

We know: —

The net potential at the rectangle center is: —

$$V = \frac{1}{4\pi\epsilon_0} \left[\frac{2q_1}{r} + \frac{-q_1}{r} + \frac{2q_1}{r} + \frac{-q_1}{r} \right] + \frac{1}{4\pi\epsilon_0} \left[\frac{4q_2}{\frac{a}{2}} + \frac{4q_2}{\frac{a}{2}} \right]$$

$$= \frac{1}{4\pi\epsilon_0} \left[\frac{4q_1}{2r} - \frac{4q_1}{2r} \right] + \frac{1}{4\pi\epsilon_0} \left[\frac{4q_2}{\frac{0.30}{2}} + \frac{4q_2}{\frac{0.30}{2}} \right]$$

P.T.O

$$V = 0 + \frac{1}{4\pi\epsilon_0} \left[\frac{4q_2}{0.105} + \frac{4q_2}{0.105} \right]$$

$$V = \frac{1}{4\pi\epsilon_0} \left[\frac{8q_2}{0.105} \right]$$

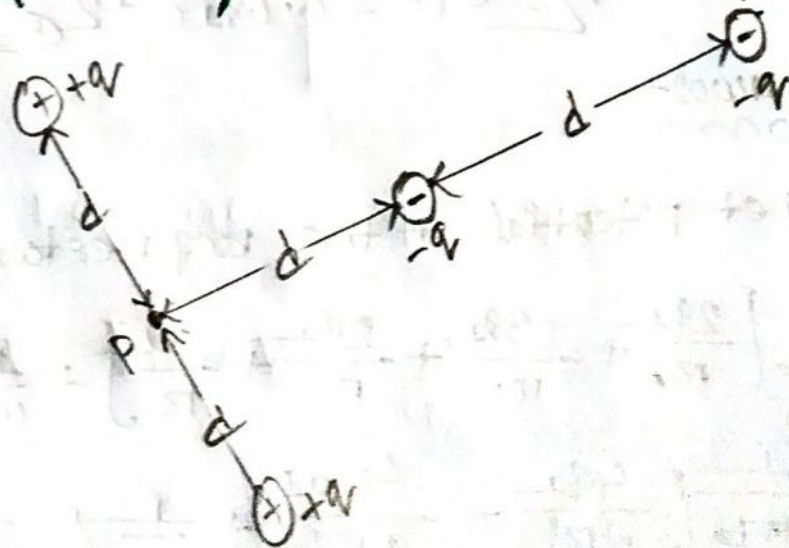
$$= 9 \times 10^9 \times \frac{8 \times 6 \times 10^{-12}}{0.105}$$

$$= 2.215 \text{ Volt}$$

Ans:-

Problem 17:-

⇒ In the adjacent figure, what is the net electric potential at point P due to the four particles if $V = 0$ at infinity, $q = 5 \text{ fC}$, and $d = 4 \text{ cm}$?



⇒ is given: —
from

$$q = 5 \text{ fC} = 5 \times 10^{-15} \text{ C}$$

$$d = 4 \text{ cm} = 0.04 \text{ m}$$

We know: —
from

The net electric potential at the point p is: —

$$V = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{d} + \frac{q}{d} + \frac{-q}{d} + \frac{-q}{2d} \right)$$

$$= \frac{1}{4\pi\epsilon_0} \times \frac{q}{d} \left(1 + 1 - 1 - \frac{1}{2} \right)$$

$$= \frac{1}{4\pi\epsilon_0} \times \frac{5 \times 10^{-15}}{0.04} \times \frac{1}{2}$$

$$= 0.75 \times 10^9 \times \frac{5 \times 10^{-15}}{2 \times 0.04}$$

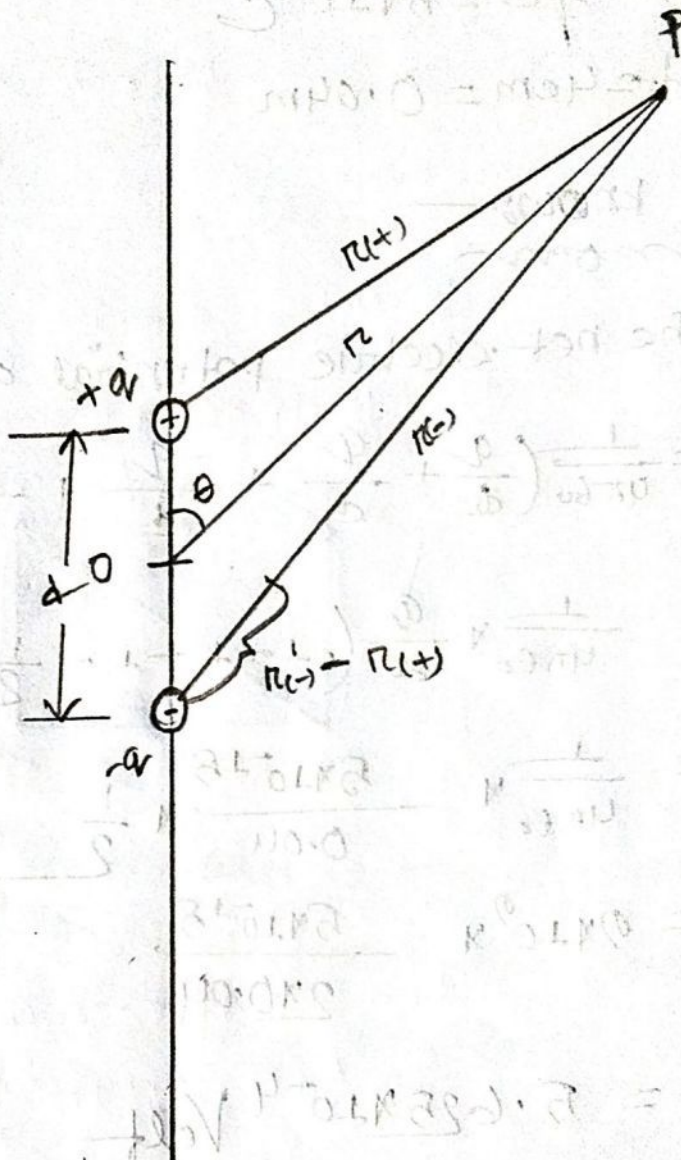
$$= 5.625 \times 10^{-4} \text{ Volt.}$$

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potential Due to an Electric Dipole:-



the net potential at P is:-

$$V = \sum_{i=1}^2 V_i = V_{(+)} + V_{(-)} = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{r(+)} + \frac{-q}{r(-)} \right]$$

$$V = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{r(+)} - \frac{1}{r(-)} \right)$$

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$$V = \frac{q}{4\pi\epsilon_0} \left(\frac{r(-) - r(+)}{r(+)*r(-)} \right) \text{ --- (iii)}$$

for dipole $d \ll r$.

So, we can write,

$$r(-) - r(+)= d \cos\theta \text{ and --- (i)}$$

$$r(-) r(+)= r^2 \text{ --- (ii)}$$

Substing these Value in number (iii) $\Rightarrow$

$$V = \frac{q}{4\pi\epsilon_0} \left(\frac{d \cos\theta}{r^2} \right)$$

$$V = \frac{1}{4\pi\epsilon_0} \left(\frac{p \cos\theta}{r^2} \right) \text{ where } p=qd$$

The vector $\vec{p}$ is directed along the dipole axis from the negative to the positive charge.

$\vec{p}$ is the magnitude of electric dipole moment

Potential due to a Line of Charge:—

Let, A thin nonconducting rod which have a positive charge.

L = Length of the rod

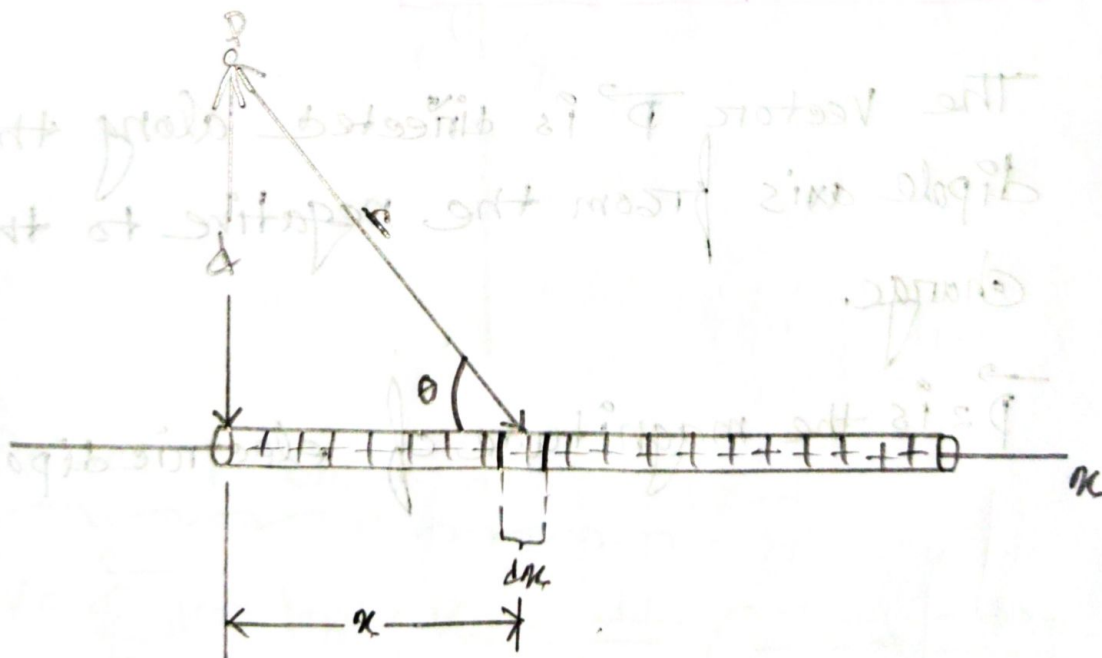
λ = Linear charge density

V = electric potential

d = distance

Now we consider a differential element $= dx$

This element of the rod has a differential charge, $dq = \lambda dx$ ———— (i)



$$r = \sqrt{d^2 + x^2}$$

$$\tan \theta = \frac{d}{x}$$

$$x = d \cot \theta$$

$$dx = -d \cot^2 \theta \cdot d\theta$$

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Let, the element as a point charge,

$$\text{So, } dV = \frac{1}{4\pi\epsilon_0} \times \frac{dq}{r^2}$$

Substitute the Value of dq and $r \Rightarrow$

$$dV = \frac{1}{4\pi\epsilon_0} \times \frac{\lambda dx}{\sqrt{d^2 + x^2}}$$

$$V = \int dV = \int_{x=0}^{x=L} \frac{1}{4\pi\epsilon_0} \times \frac{\lambda dx}{(d^2 + x^2)^{\frac{1}{2}}}$$

$$= \frac{\lambda}{4\pi\epsilon_0} \int_0^L \frac{dx}{(d^2 + x^2)^{\frac{1}{2}}}$$

$$= \frac{\lambda}{4\pi\epsilon_0} \int_0^L \frac{-d \cos \theta d\theta}{(d^2 + d^2 \cos^2 \theta)^{\frac{1}{2}}}$$

$$= \frac{\lambda}{4\pi\epsilon_0} \int_0^L -\cos \theta d\theta$$

$$= \frac{\lambda}{4\pi\epsilon_0} \times \left[\ln |\cos \theta + \cos \theta| \right]$$

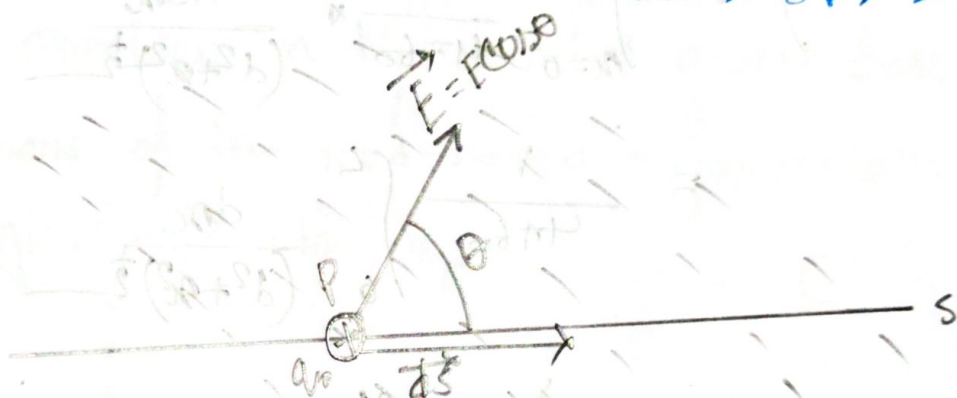
$$= \frac{\lambda}{4\pi\epsilon_0} \int_0^L \ln \left| \frac{\sqrt{d^2 + x^2}}{d} + \frac{x}{d} \right|$$

$$= \frac{\lambda}{4\pi\epsilon_0} \left[\ln \left| \frac{\sqrt{d^2 + L^2}}{d} + \frac{L}{d} \right| - \ln \left| \frac{\sqrt{d^2 - 0^2}}{d} + \frac{0}{d} \right| \right]$$

$$V = \frac{\lambda}{4\pi\epsilon_0} \ln \left| \frac{\sqrt{d^2 + L^2} + L}{d} \right|$$

V = The total potential produced by the rod at point p

Calculating the electric field from the Electric Potential:



two equipotential surface.

q_0 = positive test charge.

$\vec{r}_s$ = displacement from equipotential surface to adjacent surface.

dV = potential difference

dW = differential work.

$\vec{E}$ = Electric field.

Work done by electric potential:-

So, $dw = -q_0 dv$ — (i) | or, we know, $dv = \frac{-dw}{q_0}$

Again:-

Work done by electric field:-

$$dW = F \cdot ds \times \cos \theta$$

$$= E q_0 \times ds \times \cos \theta \quad \text{--- (ii)}$$

we know

$$E = \frac{F}{q_0}$$

$$F = E q_0$$

Now, we can write:-

$$dw = dw$$

$$-q_0 dv = q_0 E \times ds \times \cos \theta$$

$$E \cos \theta = \frac{-q_0 dv}{q_0 ds}$$

$$E \cos \theta = - \frac{dv}{ds}$$

Since $E \cos \theta$ is the component of $\vec{E}$ in the direction of $d\vec{s}$,

So,

$$E_s = - \frac{\partial V}{\partial s}$$

So, x , y , and z axes components of $\vec{E}$ at any point are,

$$E_x = - \frac{\partial V}{\partial x}; \quad E_y = - \frac{\partial V}{\partial y}; \quad E_z = - \frac{\partial V}{\partial z}$$

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Problem 2.12

mon

⇒ The ammonia molecule NH_3 has a permanent electric dipole moment equal to 1.47D , where $1\text{D} = 1\text{ debye unit} = 3.34 \times 10^{-30}\text{C.m}$. Calculate the electric potential due to an ammonia molecule at a point at a point 52.0 nm away along the axis of the dipole. (Set $V=0$ at infinity)

⇒ is given,

mon

$$p = 1.47\text{D} = 1.47 \times 3.34 \times 10^{-30}\text{C.m}$$

$$r = 52\text{ nm} = 10^{-9} \times 52$$

$Wb = Wb$

We know:

mon

$$V = \frac{1}{4\pi\epsilon_0} \times \frac{q}{r}$$

$$= \frac{1}{4\pi\epsilon_0} \times \frac{q}{r} \times \frac{r}{r}$$

$$= \frac{1}{4\pi\epsilon_0} \times \frac{q r}{r^2}$$

[Dipole moment]
 $p = qd$

$$= \frac{1}{4\pi\epsilon_0} \times \frac{p}{r^2}$$

$$= 9 \times 10^9 \times \frac{1.47 \times 3.34 \times 10^{-30}}{(52 \times 10^{-9})^2}$$

$$= 1.63 \times 10^{-5}\text{ Volt}$$

$$V_6 = \frac{1}{4\pi\epsilon_0} \times \frac{p}{r^2} = \frac{9 \times 10^9 \times 1.47 \times 3.34 \times 10^{-30}}{(52 \times 10^{-9})^2} = 1.63 \times 10^{-5}\text{ Volt}$$

Problem 368

3) The electric potential V in the space between flat Parallel plates 1 and 2 is given (in Volts) by $V = 1500x^2$, Where x (in meters) is the perpendicular distance from Plate 1. at $x_0 = 1.3 \text{ cm}$,

- (a) What is the magnitude of the electric field?
 (b) Is the field directed towards, away from plate 1?

⇒ is given

$$V = 1500x^2$$

$$x_0 = 1.3 \text{ cm} = 1.3 \times 10^{-2} \text{ m}$$

(a)

We know

$$E_x = -\frac{dV}{dx}$$

$$= -\frac{d}{dx} \times 1500x^2$$

$$= -2 \times 1500x$$

$$= -3000x$$

$$= -3000 \times 1.3 \times 10^{-2}$$

$$= -39 \text{ Vm}^{-1}$$

$$|\vec{E}| = \sqrt{(-39)^2} = 39$$

$$E = 39 \text{ Vm}^{-1}$$

Ans:—

So, $\vec{E} = E_x \hat{i} = 39(-\hat{i})$
 magnitude of $\vec{E}$ is 39

⇒ The direction of electric field towards plate 1.
 because $\vec{E} = 30(-\hat{i}) \text{ Vm}^{-1}$

Problem 372

⇒ What is the magnitude of the electric field at the point $(3\hat{i} - 2\hat{j} + 4\hat{k}) \text{ m}$ if the electric potential in the region is given by $V = 2xy^2z^2$, where V is the Volts and coordinates x, y , and z are in meters.

⇒ in given:—

$$V = 2xy^2z^2$$

$$\langle x, y, z \rangle = \langle 3, -2, 4 \rangle$$

We know:—

$$E_x = -\frac{\partial V}{\partial x} = -\frac{\partial}{\partial x} 2xy^2z^2 = -2y^2z^2$$

$$= -2 \times (-2)^2 (4)^2$$

$$= 64 \text{ Vm}^{-1}$$

$$E_y = -\frac{\partial V}{\partial y} = -\frac{\partial}{\partial y} 2xy^2z^2 = -2xz^2$$

$$= -2 \times 3 \times 4^2$$

$$= -96 \text{ Vm}^{-1}$$

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$$E_z = -\frac{\partial V}{\partial z} = -\frac{\partial}{\partial z} 2xy^2z^2 = -4m^2z$$

$$= -4 \times 3 \times 2 \times 4$$

$$= 06 \text{ Vm}^{-1}$$

$$\vec{E} = E_x \hat{i} + E_y \hat{j} + E_z \hat{k}$$

$$= 64 \hat{i} - 06 \hat{j} + 06 \hat{k}$$

$$|\vec{E}| = \sqrt{(64)^2 + (-06)^2 + (06)^2}$$

$$= 150.00 \text{ Vm}^{-1}$$

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Two Capacitors

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Capacitors —

⇒ A capacitor is an electronic component that stores electrical energy in the electric field, consisting of two conductive plates separated by a dielectric material.

Capacitance —

⇒ A capacitor consists of two isolated conductors or plates with charge $+q$ and $-q$.

A charge q and the potential difference V for the capacitor are proportional to each other.

$$q \propto V$$

$$q = CV$$

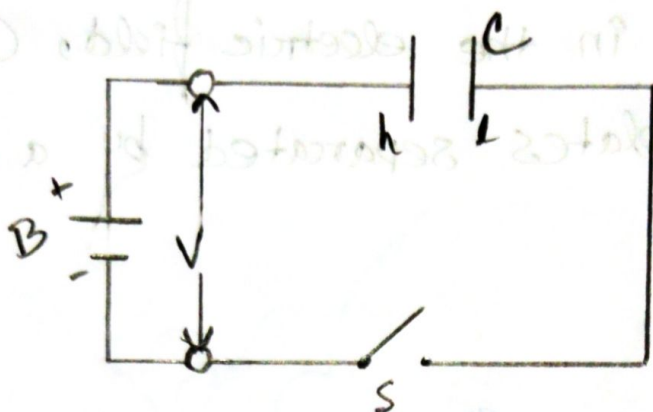
C = Capacitance of capacitor.

$$\text{So, } C = \frac{q}{V}$$

SI Unit CV^{-1} —

Common name is Farad (F);

Charging a Capacitor—



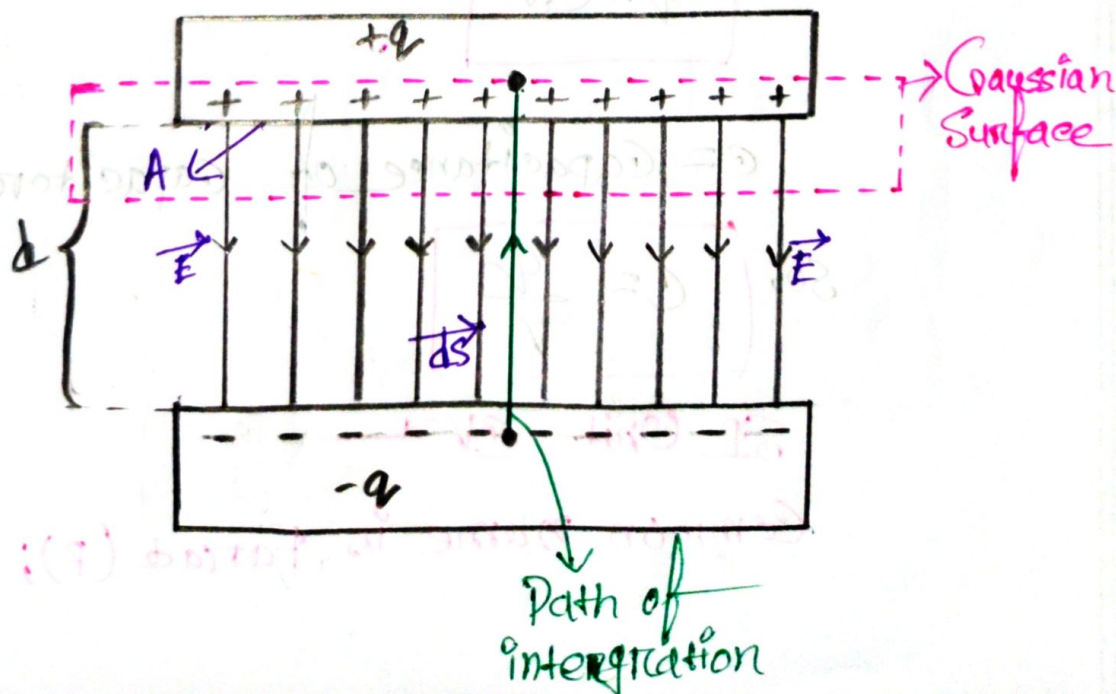
B = Battery

S = switch

h & l = plates of capacitor C.

Calculating the Capacitance:—

✶ A parallel-plate Capacitor:—



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A = Area of that part of the Gaussian surface through which there is a flux.

Applying Gauss' Law

We know:—

$$\phi = \frac{q}{\epsilon_0}$$

$$\phi \epsilon_0 = q$$

$$\epsilon_0 \oint \vec{E} \cdot d\vec{A} = q \quad [\phi = \oint \vec{E} \cdot d\vec{A}]$$

q = charge

$\oint \vec{E} \cdot d\vec{A}$ = net electric flux

$$\epsilon_0 \oint \vec{E} \cdot d\vec{A} \cdot \cos 0^\circ = q \quad \left[\begin{array}{l} \text{Since} \\ \vec{E} \text{ and } d\vec{A} \text{ are} \\ \text{Parallel} \end{array} \right]$$

$$\boxed{\epsilon_0 EA = q}$$

The potential difference between the plates of a capacitor is related to the field $\vec{E}$.

$$\begin{aligned} V_f - V_i &= - \int_i^f \vec{E} \cdot d\vec{s} \\ &= - \int_i^f E \cos 180^\circ ds \end{aligned}$$

$$V = \int_{s=d}^{s=0} E ds$$

$$[V = V_f - V_i]$$

$$V = E \int_{s=0}^{s=d} ds$$

$$\boxed{V = Ed}$$

We have:

$$C = \frac{Q}{V}$$

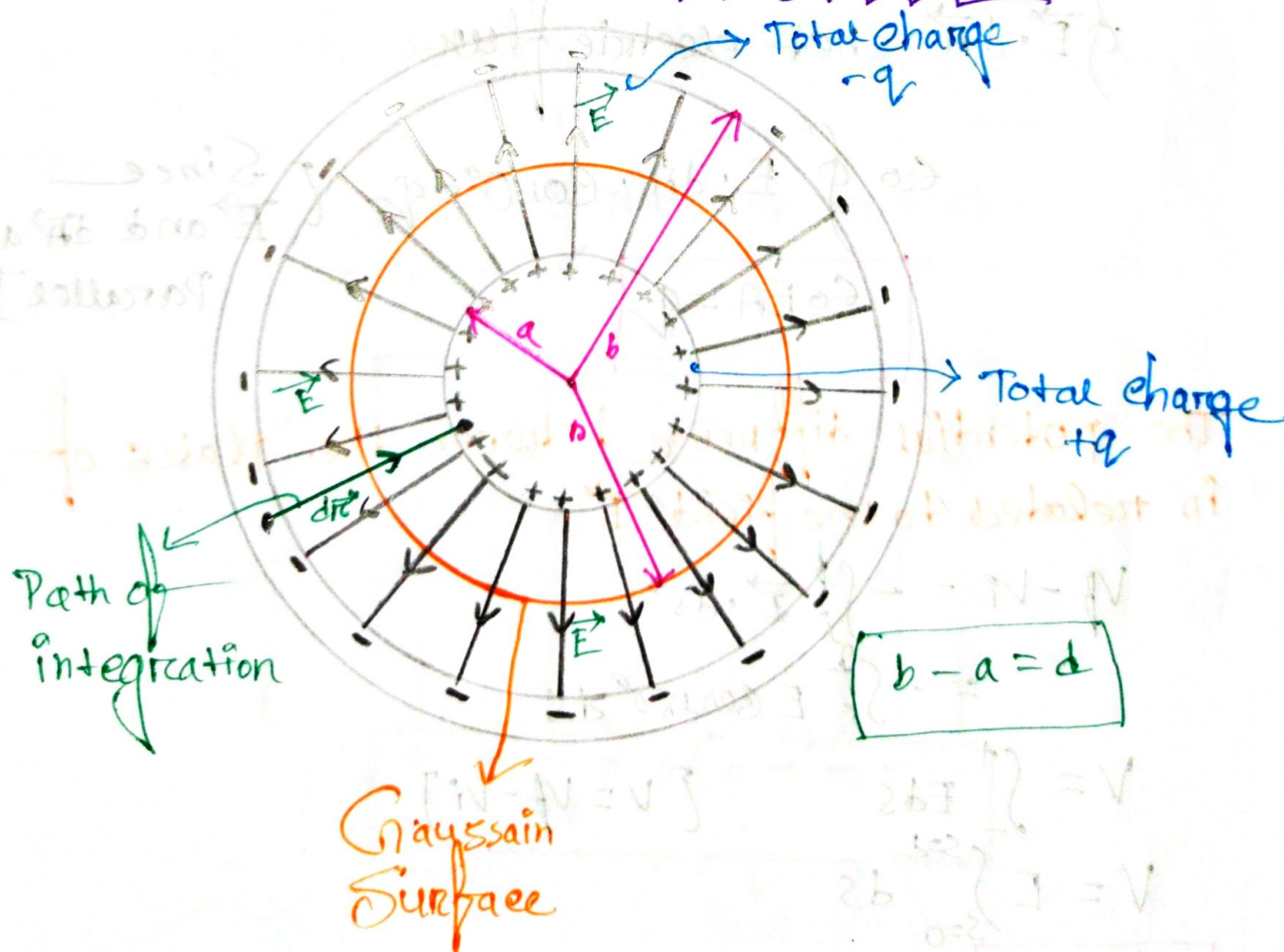
Substitute the value of V and Q

$$C = \frac{Q}{Ed}$$

$$C = \frac{\epsilon_0 EA}{Ed}$$

$$C = \frac{\epsilon_0 A}{d} = \frac{\epsilon_0 (\pi r^2)}{d} \rightarrow \text{for circular plates}$$

A Spherical Capacitor



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 r_0 = sphere of radius a & b = two concentric spherical shells of radiiApplying Gauss' Law:We can find:

$$\epsilon_0 EA = q$$

$$\epsilon_0 E(4\pi r^2) = q$$

$$E = \frac{q}{4\pi\epsilon_0 r^2} \quad \text{--- (1)}$$

where,

 $A = 4\pi r^2$ is the area of the spherical Gaussian Surface.We know:

$$V = \int_{-}^{+} E ds = - \int_{-}^{+} E dr \quad \left[\text{Since } ds = -dr \right]$$

$$V = - \frac{q}{4\pi\epsilon_0} \int_{r=b}^{r=a} \frac{dr}{r^2}$$

$$= - \frac{q}{4\pi\epsilon_0} \left(\left| -\frac{1}{r} \right|_b^a \right)$$

$$= - \frac{q}{4\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{b} \right)$$

$$= - \frac{q}{4\pi\epsilon_0} \times \frac{b-a}{ab}$$

$$= - \frac{q(b-a)}{4\pi\epsilon_0 ab} \quad \text{--- (11)}$$

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Substitute the Value of V in the equation of

$$C = \frac{q}{V}$$

$$= \frac{q}{\frac{q(b-a)}{4\pi\epsilon_0 ab}}$$

$$= q \times \frac{4\pi\epsilon_0 ab}{q(b-a)}$$

$$C = \frac{4\pi\epsilon_0 ab}{(b-a)}$$

$$\therefore C = \frac{4\pi\epsilon_0 ab}{(b-a)}$$

An isolated sphere:-

⇒ We can assign a capacitance to a single isolated spherical conductor of radius R by assuming that the "missing plate" is a conducting sphere of infinite radius.

We know:-
from

$$C = \frac{4\pi\epsilon_0 ab}{b-a}$$

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$$\Rightarrow C = \frac{\frac{4\pi\epsilon_0 ab}{b-a}}{\frac{b}{b-a}}$$

[Dividing both by b]

$$= \frac{4\pi\epsilon_0 a}{1 - \frac{a}{b}}$$

$$= \frac{4\pi\epsilon_0 a}{1 - \frac{a}{\infty}}$$

[$b = \infty$]

$$= \frac{4\pi\epsilon_0 a}{1 - 0}$$

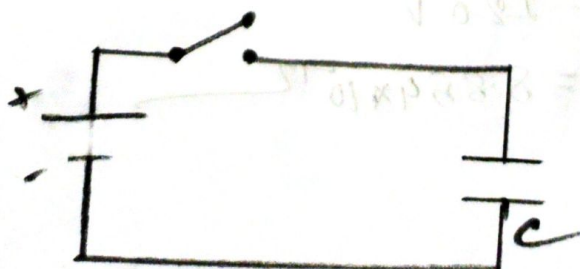
$$= \frac{4\pi\epsilon_0 a}{1 - 0}$$

$$\therefore C = 4\pi\epsilon_0 a$$

→ This is the capacitance for isolated sphere.

Problem 2

⇒ The Capacitor in the adjacent Fig. has a capacitance of $25\mu F$ and is initially uncharged. The battery provides a potential difference of $120 V$. After switch S is closed, how much charge will pass through it?



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⇒ is given: —

$$C = 25 \mu\text{F} = 25 \times 10^{-6} \text{ F}$$

$$V = 120 \text{ V}$$

We know: —
mom

$$C = \frac{Q}{V}$$

$$\begin{aligned} Q &= CV \\ &= 25 \times 10^{-6} \times 120 \\ &= 3 \times 10^{-3} \text{ C} \end{aligned}$$

Ans

Problem: 3
mom

⇒ A parallel-plate capacitor has circular plates of 8.20 cm radius and 1.30 mm separation.

(a) Calculate the capacitance.

(b) Find the charge for a potential difference of 120 V.

⇒ is given: —
mom

$$r = 8.20 \text{ cm} = 8.20 \times 10^{-2} \text{ m}$$

$$d = 1.30 \text{ mm} = 1.30 \times 10^{-3} \text{ m}$$

$$V = 120 \text{ V}$$

$$\epsilon_0 = 8.854 \times 10^{-12}$$

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(a)

We know:
mom

$$C = \frac{\epsilon_0 (\pi r^2)}{d}$$

$$= \frac{8.854 \times 10^{-12} \times (\pi \times (8.2 \times 10^{-2})^2)}{1.30 \times 10^{-3}}$$

$$= 1.43 \times 10^{-10} \text{ F}$$

Ans:

(b)

We know:
mom

$$C = \frac{Q}{V}$$

$$Q = CV$$

$$= 1.43 \times 10^{-10} \times 120$$

$$= 1.72 \times 10^{-8} \text{ C}$$

Problem: 40-

⇒ The plates of a spherical capacitor have radii 38 mm and 40 mm.

(a) Calculate the capacitance.

(b) What must be the plate area of a parallel-plate capacitor with the same plate separation and capacitance.

⇒ is given: _____

$$a = 38 \text{ mm} = 38 \times 10^{-3} \text{ m}$$

$$b = 40 \text{ mm} = 40 \times 10^{-3} \text{ m}$$

(a)

We know:-

$$C = \frac{4\pi\epsilon_0 ab}{b-a}$$

$$= \frac{4 \times 8.854 \times 10^{-12} \times \pi \times 38 \times 10^{-3} \times 40 \times 10^{-3}}{(40 - 38) \times 10^{-3}}$$

$$= 8.45 \times 10^{-11} \text{ F}$$

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(b)

$$d = b - a = 2 \times 10^{-3} \text{ m}$$

we know:-
mom

for a parallel plate capacitor:-

$$C = \frac{\epsilon_0 A}{d}$$

$$A = \frac{Cd}{\epsilon_0}$$

$$= \frac{8.45 \times 10^{-11} \times 2 \times 10^{-3}}{8.854 \times 10^{-12}}$$

$$= 10.087 \times 10^{-3} \text{ m}^2$$

Ans:-

Problem 6:-

mom
You have two flat metal plates, each of area 1 m^2 with which to construct a parallel plate capacitor.

(a) if the capacitance of the device is to be 1 F what must be the separation between the plates.

(b) Could this capacitor actually be constructed?

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⇒ is given: —

$$A = 1 \text{ m}^2$$

$$C = 1 \text{ F}$$

$$\epsilon_0 = 8.854 \times 10^{-12}$$

(a)

we know —

$$C = \frac{\epsilon_0 A}{d}$$

$$d = \frac{\epsilon_0 A}{C}$$

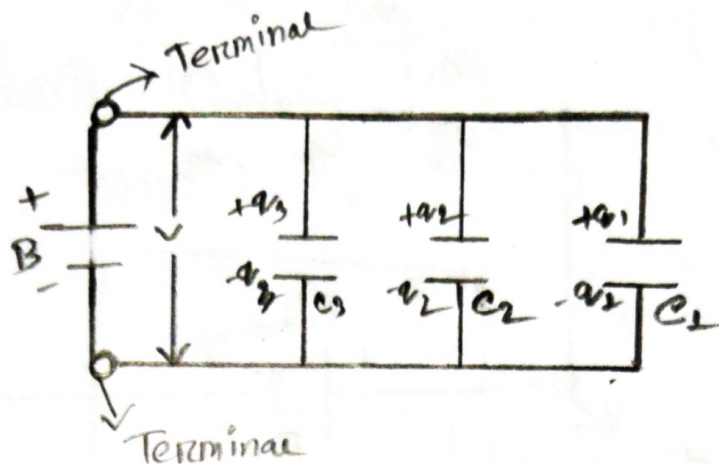
$$= \frac{8.854 \times 10^{-12} \times 1}{1}$$

$$= 8.854 \times 10^{-12} \text{ m}$$

(b)

It is not possible to construct a capacitor by the separation distance, $d = 8.854 \times 10^{-12} \text{ m}$ because d value is less than the minimum size of atom.

Capacitors in parallel Combination:-



Charge on each capacitor:-

$$q_1 = C_1 V$$

$$q_2 = C_2 V$$

$$q_3 = C_3 V$$

$$q_1 = C_1 V$$

So, The total charge on parallel combination is:-

$$q = q_1 + q_2 + q_3$$

$$q = (C_1 + C_2 + C_3) V$$

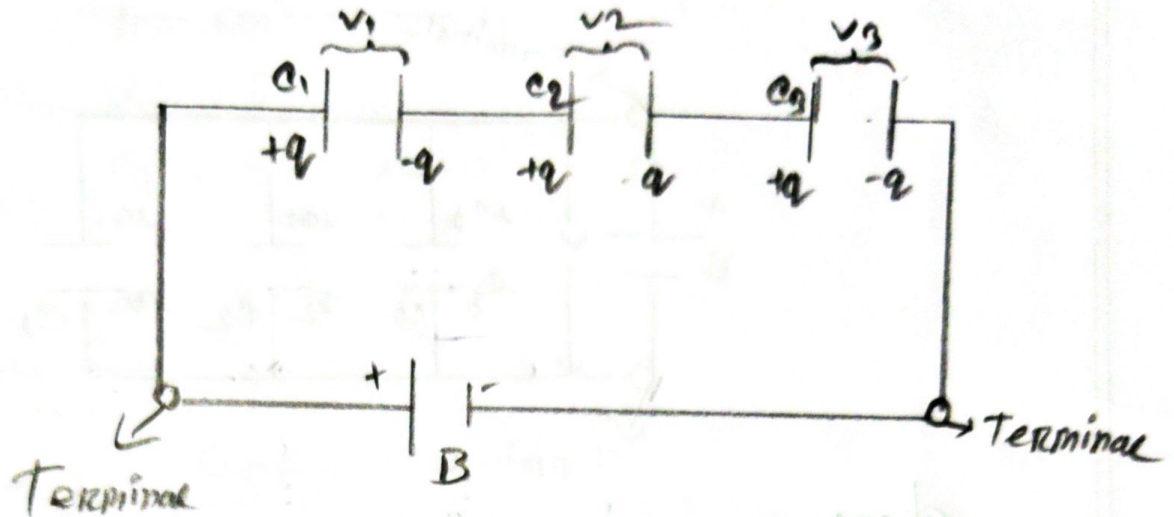
Now:- The equivalent capacitance is:-

$$C_{eq} = \frac{q}{V}$$

$$C_{eq} = \frac{(C_1 + C_2 + C_3) V}{V}$$

$$C_{eq} = C_1 + C_2 + C_3$$

#Capacitors in Series Combination—



Potential difference of each actual capacitor:—

$$V_1 = \frac{q}{C_1} ; V_2 = \frac{q}{C_2} ; V_3 = \frac{q}{C_3}$$

So, the total potential difference $= V$ is;

$$\begin{aligned} V &= V_1 + V_2 + V_3 \\ &= q \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \right) \end{aligned}$$

Now: The equivalent capacitance is:—

$$\begin{aligned} C_{eq} &= \frac{q}{V} \\ &= \frac{q}{q \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \right)} \end{aligned}$$

$$\boxed{\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}}$$

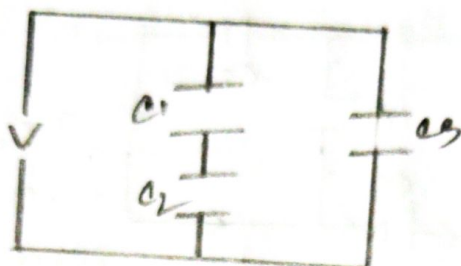
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Problem 10:

⇒ In the adjacent figure, find the equivalent capacitance of the combination. Assume that $C_1 = 10 \mu F$, $C_2 = 5 \mu F$, $C_3 = 4 \mu F$



⇒ is given

$$C_1 = 10 \mu F$$

$$C_2 = 5 \mu F$$

$$C_3 = 4 \mu F$$

here, C_1 and C_2 are in Series

$$\begin{aligned} \frac{1}{C_s} &= \frac{1}{C_1} + \frac{1}{C_2} \\ &= \frac{1}{10} + \frac{1}{5} \\ &= \frac{3}{10} \end{aligned}$$

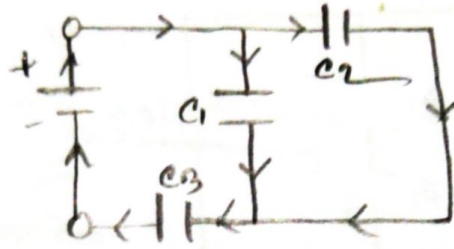
$$C_s = \frac{10}{3} \mu F$$

And C_s and C_3 are in parallel

$$\begin{aligned} C_{eq} &= C_s + C_3 \\ &= \frac{10}{3} + 4 \\ &= \frac{22}{3} \mu F \\ &= 7.33 \times 10^{-6} F \end{aligned}$$

Problem 128

⇒ In the adjacent figure, find the equivalent capacitance of the combination. Assume that $C_1 = 10 \mu F$, $C_2 = 5 \mu F$, $C_3 = 4 \mu F$



⇒ is given —

$$C_1 = 10 \mu F$$

$$C_2 = 5 \mu F$$

$$C_3 = 4 \mu F$$

here, C_1 and C_3 are in Series —

$$\frac{1}{C_{eqs}} = \frac{1}{C_1} + \frac{1}{C_3}$$

$$= \frac{1}{10} + \frac{1}{4}$$

$$= \frac{7}{20}$$

$$\therefore C_{eqs} = \frac{20}{7} \mu F$$

Now, C_{eqs} and C_2 are in Parallel —

$$\therefore C_{eq} = C_{eqs} + C_2$$

$$= \frac{20}{7} + 5$$

$$= \frac{55}{7} \mu F$$

$$= 7.85 \times 10^{-6} F$$

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⇒ is given —
mom

$$C_1 = 10 \mu F$$

$$C_2 = 5 \mu F$$

$$C_3 = 4 \mu F$$

Here, C_1 and C_2 are in parallel —

$$\begin{aligned} C_{eqp} &= C_1 + C_2 \\ &= 10 + 5 \\ &= 15 \mu F \end{aligned}$$

Now — C_{eqp} and C_3 are in Series: —

$$\frac{1}{C_{eqs}} = \frac{1}{C_{eqp}} + \frac{1}{C_3}$$

$$\begin{aligned} &= \frac{1}{15} + \frac{1}{4} \\ &= \frac{10}{60} \end{aligned}$$

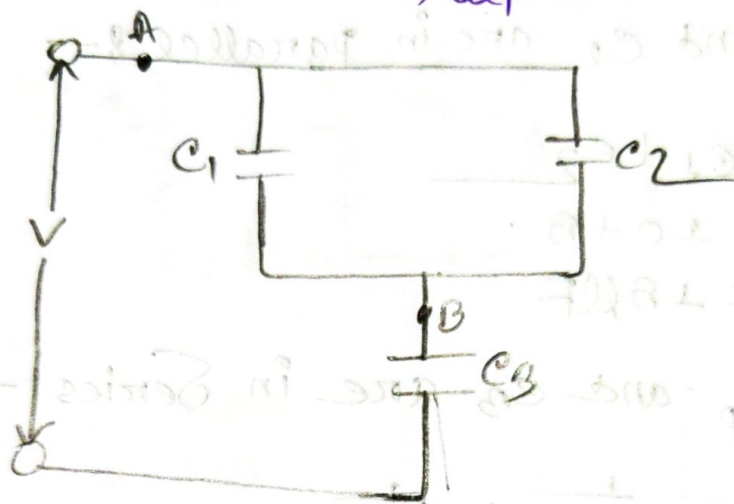
$$\therefore C_{eqs} = \frac{60}{10} \mu F$$

$$= 3.157 \times 10^{-6} F$$

Ans: —

Problem 8.5.02

⇒ Find the equivalent capacitance for the combination of capacitances shown in figure, Assume $C_1 = 12 \mu\text{F}$, $C_2 = 5.30 \mu\text{F}$, $C_3 = 4.5 \mu\text{F}$



⇒ here,

$$C_1 = 12 \mu\text{F}$$

$$C_2 = 5.30 \mu\text{F}$$

$$C_3 = 4.5 \mu\text{F}$$

here, C_1 and C_2 are in parallel:-

$$C_{12} = 12 + 5.30$$

$$= 17.3 \mu\text{F}$$

Now:- C_{12} and C_3 are in Series:-

$$\frac{1}{C_{123}} = \frac{1}{C_{12}} + \frac{1}{C_3}$$

$$= \frac{1}{17.3} + \frac{1}{4.5}$$

$$= \frac{436}{1557}$$

$$C_{123} = 3.57 \mu\text{F}$$

Ans:-

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#Energy Stored in an Electric Field:—

⇒ Suppose that,

at a given instant, a charge q' has been transferred from one plate of a capacitor to other. The potential difference V' between the plates at that instant will be $\frac{q'}{C}$.

If the extra increment of charge dq' is then transferred, the increment of work required will be

$$dW = V' dq'$$

$$dW = \frac{q'}{C} \times dq'$$

The work required to bring the total capacitor charge up to a final value q is

$$W = \int_{q'=0}^{q'=q} dW = \frac{1}{C} \int_0^q q' dq' = \frac{q^2}{2C}$$

The Work Stored as potential energy U in the Capacitor,

$$U = W = \frac{q^2}{2C} = \frac{(eV)^2}{2C} = \frac{e^2 V^2}{2C} = \frac{1}{2} eV^2$$

$$\therefore U = \frac{q^2}{2C} = \frac{1}{2} eV^2$$